

from the 1970s until the 1990s introduced genetic modifications. For example, they engineered plague bacteria to be resistant to sixteen different antibiotic drugs and to produce a toxin that adds paralysis to the list of its effects. International efforts to ban bio-weapon development in the wake of the attacks on the World Trade Center in New York City on September 11, 2001, might put an end to this subversion of biotechnology. **SEE ALSO** AGRICULTURAL BIOTECHNOLOGY; CLONING: ETHICAL ISSUES; GENE THERAPY: ETHICAL ISSUES; GENETIC DISCRIMINATION; GENETIC TESTING: ETHICAL ISSUES; METABOLIC DISEASE; REPRODUCTIVE TECHNOLOGY: ETHICAL ISSUES; TRANSGENIC ORGANISMS: ETHICAL ISSUES.

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Biotechnology and Genetic Engineering, History of

The term "biotechnology" dates from 1919, when the Hungarian engineer Karl Ereky first used it to mean "any product produced from raw materials with the aid of living organisms." Using the term in its broadest sense, biotechnology can be traced to prehistoric times, when hunter-gatherers began to settle down, plant crops, and breed animals for food. Ancient civilizations even found that they could use microorganisms to make useful products, although, of course, they had no idea that it was microbes that were the active agents. About B.C.E. 7000, the Sumarians and Babylonians discovered how to use yeast to make beer, and wine-making dates from biblical times. In about B.C.E. 4000, the Egyptians found that the addition of yeast produced a light, fluffy bread instead of a thin, hard wafer. At the same time, the Chinese were adding bacteria to milk to produce yogurt.

Genetic Engineering versus Biotechnology

For many, the term "biotechnology" is often equated with the manipulation of genes, but, as Ereky's definition suggests, this is only one aspect of biotechnology. For the more specific technique of gene manipulation, the term "genetic engineering" is more appropriate. Genetic engineering dates from the 1970s. At that time molecular biologists devised methods to isolate, identify, and clone genes as well as to mutate, manipulate, and insert them into other species. One of the key elements in such research was the discovery of **restriction enzymes**. These enzymes are able to cleave DNA at a limited number of sequence-specific sites and often leave "sticky ends." Isolated DNA from any organism could be cleaved with a restriction enzyme and then mixed with a preparation of a **vector** that had been cleaved with the same restriction endonuclease. By virtue of the "sticky ends," a **hybrid**

restriction enzymes
enzymes that cut DNA
at a particular sequence

vector carrier

hybrid combination of
two different types



Egyptian artwork, dating from between B.C.E. 1550 and 1295, depicts the harvest of the grapes and subsequent counting of the jars of wine. This art suggests that ancient civilizations fermented grape juice to make wine, establishing the basics of a process still used in wineries today.

molecule could be created that contained the gene of interest, which could then be inserted into such a cloning vector. The importance of restriction endonucleases was recognized in 1978 by the awarding of the Nobel Prize in physiology or medicine to Werner Arber, Daniel Nathans, and Hamilton Smith for their discovery of these enzymes.

Further Advances and Ethical Concerns

The first experiment to combine different DNA molecules was performed in 1972 in the laboratory of Paul Berg (who shared the 1980 Nobel Prize in chemistry for this work). The following year Stanley Cohen and Herbert Boyer combined some viral DNA and bacterial DNA in a **plasmid** to create the first recombinant DNA organism.

Realizing the potential dangers of moving genes from one organism to another, approximately ninety prominent scientists, whose laboratories were poised to start cloning experiments, met in 1975 at the Asilomar Conference Center in California to discuss the potential dangers of gene manipulation. This meeting, wherein scientists recognized and openly discussed the ramifications and potential dangers of their research before that research was actually begun, was unprecedented. The result of the Asilomar Conference was to call for and agree upon a one-year moratorium before any cloning experiments were to be done. This provided time to develop guidelines for the physical and biological isolation of recombinant organisms, to ensure that they not escape into the environment, and, if they did, to make sure that they would be so weakened as not to survive competition with naturally occurring organisms. By 1976, then, gene cloning was in full swing around the world.

plasmid a small ring of DNA found in many bacteria





recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms

hormone molecule released by one cell to influence another

Key Technical Developments

Advances in biotechnology were marked by the development of key research techniques. In 1976, Herbert Boyer and Robert Swanson founded Genentech, the first biotechnology company to use **recombinant DNA** technology in developing commercially useful products such as drugs. The year 1977 is considered the “dawn of modern biotechnology,” for it was in that year that the first human protein was cloned and manufactured using genetic engineering technology: Genentech reported the cloning of the human **hormone** somatostatin. This year was also important for the development of the technique of DNA sequencing, achieved by Fred Sanger and Walter Gilbert (who, with Paul Berg, shared the 1980 Nobel Prize in chemistry).

In 1978 Genentech was able to isolate the gene for human insulin and begin clinical trials that resulted in the approval and marketing of the first genetically engineered drug for human use. This was a major accomplishment. Diabetes, the seventh leading cause of death in the United States, affects millions of Americans. In the past, insulin was extracted from the pancreases of cows or pigs, then used to treat diabetics. Although insulin from these species is very similar to human insulin and was effective in humans, the small differences between human and animal insulin were enough to cause problems for some patients. Often patients developed immunological reactions to the foreign protein, reducing its effectiveness. With the availability of genetically engineered human insulin, these problems were eliminated.

Patents and the Rise of Biotechnology Companies

In 1980 the U.S. Supreme Court provided an important incentive for the development of biotechnology companies. In the case of *Diamond v. Chakrabarty*, the court ruled that biological materials may be patented. Thus, private companies could look forward to making substantial profits from therapies that they developed through genetic engineering techniques.

Among the new companies to take advantage of the court ruling was the Chiron corporation, which cloned the protein that formed the outer coat of the human hepatitis B virus. This protein, which could now be produced without the virus that it normally enclosed, provided the material for the development of the first human vaccine using recombinant DNA technology. The hepatitis **vaccine** has been available since 1987.

vaccine protective antibodies

In the same year, the Food and Drug Administration (FDA) approved Genentech’s drug tPA (tissue plasminogen activator). This is a human blood protein that helps to dissolve fibrin, the major protein involved in forming blood clots at the site of an injury. After the healing process is complete and clotting is no longer required at the site of the injury, the body normally releases tPA to activate an enzyme called plasmin, which dissolves fibrin. However, it was discovered that tPA could also be used as a powerful drug in the treatment of certain heart attacks. Sometimes a blood clot forms spontaneously in the body. If the clot forms or lodges in the coronary arteries of the heart, the clot blocks blood flow to the heart muscle, resulting in what is commonly called a heart attack. If tPA is given to such patients within four hours of onset, the recovery is truly remarkable. Such patients are able to leave the hospital the next day with little or no after-

effects of the heart attack. A patient not treated with tPA often remains in the hospital for a week or longer and can not resume normal activities until after a long recovery period. The drug has subsequently been approved for use with patients suffering a stroke from a blood clot in the brain with similar success.

Biotechnology has also been successful in development of other useful products. Today many laundry detergents contain proteases, enzymes that remove stains by digesting the protein components of the stain. However, such enzymes are inactivated by bleach. In 1988 the biotechnology company Genecor received approval for a bleach-resistant protease. This had been accomplished by isolating the gene for protease and then, using site-directed mutagenesis, changing the gene such that the corresponding protein was no longer sensitive to inactivation by bleach.

Biotechnology has also made a great impact in agriculture. The first genetically engineered plant was patented in 1983. The first genetically engineered food was produced by a company called Calgene, in 1987. Calgene, now a part of Monsanto, produced a tomato that could be ripened on the vine and transported ripe to market. Tomatoes are normally shipped green to market and left to ripen at their destination because they are easily bruised and damaged if shipped when fully ripe. Today there is a new “green revolution” under way, in which genetically modified food will provide greater nourishment and higher yields, while simultaneously reducing the use of fertilizers and herbicides. Although there is considerable controversy surrounding these foods (sometimes referred to as “Frankenfood”), there have been no documented cases of anyone being hurt by eating them. In 1990 the biotech firm GenPharm created a transgenic dairy cow into which the genes for human milk proteins were inserted. The milk from such cows will be used for producing infant formula.

Biotechnology and the Law

Biotechnology has also made important contributions to the field of law. Most notably, scientists have developed exquisitely sensitive methods for identifying DNA. Indeed, with the invention of the **polymerase** chain reaction in 1988, enough DNA can be extracted from a drop of blood, a tiny shred of skin, a single hair, or a small semen sample to identify the individual from whom it originated. Such “genetic fingerprinting” was developed in 1984 and first used in a trial in 1985.

Perhaps the most famous case involving DNA-based evidence was the O. J. Simpson murder trial in 1995. During the 1990s however, genetic evidence in the courtroom became commonplace and accepted by trial lawyers, judges, and juries alike. In fact, several innocent people were released from prison as a result of the reexamination of evidence using DNA fingerprinting.

The Human Genome Project

In 1990 molecular biologists around the world began working on what ranks as perhaps the greatest achievement of biotechnology, the Human Genome Project, in which the more than 3 billion **nucleotides** of DNA in the human nucleus were ultimately sequenced. Although DNA sequencing began in 1977, it was the development in the 1990s of automated DNA sequencers and powerful computers to store and analyze the data that made this project

polymerase enzyme complex that synthesizes DNA or RNA from individual nucleotides

nucleotides the building blocks of RNA or DNA



feasible. The first draft of the human genome was completed in 2001. With the complete sequence available, scientists will be able to “mine the genome” to find important gene products and to design specific drugs to target gene product. The twenty-first century will see tremendous new advances using biotechnology. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; BIOINFORMATICS; BIOTECHNOLOGY; BIOTECHNOLOGY ENTREPRENEUR; CLONING GENES; CLONING ORGANISMS; GENETICALLY MODIFIED FOODS; GENOMICS INDUSTRY; HUMAN GENOME PROJECT; MUTAGENESIS; PATENTING GENES; PLANT GENETIC ENGINEER; RESTRICTION ENZYMES; SANGER, FRED.

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Birth Defects

A birth defect is an anomaly that is congenital, or present from birth. Birth defects are the leading cause of infant mortality, causing 22 percent of all infant deaths. Approximately 3 to 4 percent of all live births are affected by a birth defect; the causes of most of them are unknown. Some birth defects are considered to be physical, while others are thought of as functional. Physical birth defects result in the malformation of a physical organ or limb, whereas functional birth defects are those that cause primarily functional, rather than physical, problems. Functional birth defects include mental retardation, congenital hearing loss, early-onset vision impairment, and numerous other health concerns. They may be caused by single gene mutations, or they may be due to polygenic or multifactorial inheritance.

Birth defects may be found in isolation or they may occur in combination in one child, as part of a larger syndrome. A syndrome is usually defined as the presence of three or more birth defects due to one underlying cause, that characterize a particular disease or condition. An example of this is Down syndrome.

Various Causes, Various Treatments

There are several **etiologies** of birth defects, including single gene mutations, polygenic and multifactorial conditions, chromosomal abnormalities, and teratogens, which cause growth or developmental abnormalities. These etiologies may cause both physical and functional birth defects. The diagnosis, treatment, and management of birth defects often involves a team of professionals and specialists. Among these specialists are clinical geneticists, medical geneticists, and genetic counselors. These are all genetic service providers who are trained to help make a diagnosis and identify whether a birth defect is isolated or part of a syndrome. An accurate diagnosis is of utmost importance in being able to treat the condition and anticipate any future health concerns that may arise.

etiologies causations
of disease

LEADING CATEGORIES OF BIRTH DEFECTS

Birth Defects	Estimated Incidence
Structural/Metabolic	
Heart and circulation	1 in 115 births
Muscles and skeleton	1 in 130 births
Club foot	1 in 735 births
Cleft lip/palate	1 in 930 births
Genital and urinary tract	1 in 135 births
Nervous system and eye	1 in 235 births
Anencephaly	1 in 8,000 births
Spina bifida	1 in 2,000 births
Chromosomal syndromes	1 in 600 births
Down syndrome (Trisomy 21)	1 in 900 births
Respiratory tract	1 in 900 births
Metabolic disorders	1 in 3,500 births
PKU	1 in 12,000 births
Congenital Infections	
Congenital syphilis	1 in 2,000 births
Congenital HIV infection	1 in 2,700 births
Congenital rubella syndrome	1 in 100,000 births
Other	
Rh disease	1 in 1,400 births
Fetal alcohol syndrome	1 in 1,000 births

Adapted from "Leading Categories of Birth Defects." March of Dimes Perinatal Data Center, 2000.

Single-Gene Mutations

Single-gene mutations are defects or changes in genes that may be passed on from generation to generation. The most common forms of inheritance include **autosomal** dominant, autosomal recessive, X-linked recessive, and new dominant mutations. Single-gene mutations may result in both structural and functional birth defects. These defects may be found in isolation or as part of a known syndrome and may display **phenotypic** variation, in which the same mutant gene leads to variable clinical problems. This often occurs with dominant mutations such as Marfan syndrome. A genetic disorder may also display reduced penetrance, wherein not everyone with the genetic mutation will have the disorder or trait. These are also known as "silent" disorders.

Autosomal Dominant Disorders

An example of an autosomal dominant disorder is achondroplasia, the most common form of short-limbed dwarfism in humans. Achondroplasia displays complete penetrance (everyone with the genetic defect also has the disorder), and it occurs in 1 out of 25,000 births. Most cases are **sporadic** rather than inherited.

Achondroplasia is a growth disorder caused by a mutation of the gene that encodes the fibroblast growth factor receptor 3 (*FGFR3*), and it is

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

phenotypic related to the observable characteristics of an organism

sporadic caused by new mutations

characterized by short limbs, malformed hands, a disproportionately large head, and abnormal facial features. Medical problems are due to abnormally configured bones and related structures, leading to hydrocephalus, problems of the spine, frequent sinus and ear infections, and orthopedic problems. If one parent is affected with achondroplasia, there is a 50 percent risk that an offspring will also be affected. If both parents are affected, there is a 25 percent chance that an offspring will inherit two gene copies and develop severe, life-threatening features. Two known mutations in *FGFR3* account for 98 percent of all achondroplasia cases. This makes early identification, even prenatal diagnosis, relatively easy.

Another autosomal dominant condition is Marfan syndrome, which results from a defect in the synthesis, secretion, or utilization of the protein fibrillin, an important component of connective tissue throughout the body. The gene for Marfan syndrome is fibrillin 1 (*FBR1*). Marfan syndrome features are variable, including cardiovascular, skeletal, and ocular defects.

Marfan syndrome's most serious medical complication is the risk of sudden death from aortic dissection, a tear in the inner wall of the major artery leading from the heart. Approximately 75 percent of individuals with Marfan syndrome have a family history of the disease, with the rest occurring as new mutations. Because the syndrome is autosomally dominant, affected individuals have a 50 percent risk of passing the mutated gene to their offspring. The condition has full penetrance; therefore, all individuals who inherit this mutation will express some features of Marfan syndrome.

Functional Birth Defects

Fragile X syndrome is the most common cause of inherited mental retardation, occurring in one out of 1,000 births. It is caused by expansion of a "triplet repeat" section of nucleotides in the *FMR-1* gene on the X chromosome. Triplet repeats are three-**base-pair** sequences in a gene that are abnormally repeated, sometimes dozens or even hundreds of times, causing abnormal protein sequence and structure. Because it is carried on the X chromosome, it affects males more often than females. In males, an *FMR-1* gene with greater than 200 repeats is always associated with the syndrome. Inactivated *FMR-1* gene causes impaired mental function. The *FMR-1* protein is thought to help shape the connections between **neurons** that underlie learning and memory. Affected individuals may also have large testes, abnormal facial features, seizures, and emotional and behavior problems. DNA testing allows for detection of carriers as well as affected individuals, enabling the use of genetic counseling and prenatal testing.

Another common functional genetic disorder is hearing loss, which can result from a defect in any one of more than fifty different genes. One gene, connexin 26, may be responsible for a large portion of inherited hearing loss. Some forms of congenital hearing loss may be due to prenatal exposure to infectious agents such as rubella. Genetic screening or screening for hearing loss at birth may be the most important test for hearing impairment yet to be developed, as early recognition and treatment can lead to dramatic improvements in hearing and, consequently, in the development of language in early childhood.

Other genetic disorders that cause functional birth defects include those involved in various aspects of the immune system. The most severe form

base pair two nucleotides (either DNA or RNA) linked by weak bonds

neurons brain cells

is severe combined immune deficiency (SCID), in which a major type of immune cell, the lymphocyte, is absent. People with SCID suffer life-threatening infections beginning in infancy and may require complete physical isolation. This was the case for David Vetter, who became known to the world as the “bubble boy.”

Multifactorial and Polygenic Inheritance

Many traits and diseases are caused by the interaction of inherited genes and the environment. These are known as “multifactorial” traits. While all genes interact with the environment, the impact of the environment in multifactorial traits and diseases is usually greater than in single-gene traits and diseases. Prenatal environmental influences are inevitably filtered through the maternal-placental system and include factors such as infections, drugs, tobacco or alcohol use, diabetes, and industrial toxins.

Polygenic traits and diseases are due to the cumulative effect of multiple genes, working together. Many congenital birth defects are thought to be multifactorial, such as pyloric stenosis (narrowing of the passage from stomach to intestine), cleft lip and palate, clubfoot, and neural tube defects. When found as isolated birth defects, these conditions are thought to be explained by a “multifactorial threshold model.”

The multifactorial threshold model assumes the gene defects for multifactorial traits are normally distributed within the population. This means that almost everyone has some genes involved with these conditions, with most individuals having too few of them to cause disease. Individuals will not become affected with the condition unless they have a genetic liability that is significant enough to push them past the threshold, moving them out of the unaffected range and into the affected range (Figure 1).

Examples of Multifactorial and Polygenic Effects

Cleft lip with or without cleft palate (CL/P) is a heterogeneous disorder (those children affected may have somewhat different abnormalities) occurring in 1 out of 1,000 births. Some CL/P cases occur as isolated birth defects, while others occur as part of a larger syndrome. The majority of CL/P cases are associated with multifactorial inheritance. The risk to relatives of affected individuals can be anywhere from 0.5 to 15 percent, depending on the severity of the clefting and the degree of relationship to the affected individual, with risks highest for first-degree relatives. Some unique cases of CL/P may be associated with genetic syndromes that are due to single-gene mutations or chromosomal abnormalities.

Clubfoot is another primarily multifactorial defect and occurs in 1 out of 10,000 Caucasian newborns. The estimated risk to relatives of inheriting this defect is between 2 to 20 percent, depending upon the family history. Clubfoot can also have genetic causes such as chromosomal abnormalities or single-gene disorders, or it may have an environmental origin, such as problems caused by amniotic fluid or structural abnormalities of the uterus that restrict fetal growth and mobility. Clubfoot can also be due to autosomal recessive as well as autosomal dominant inheritance, and it may also occur as part of a larger syndrome.

Another class of multifactorial disorders is known as neural tube defects (NTDs). The neural tube is the embryonic structure that develops into the

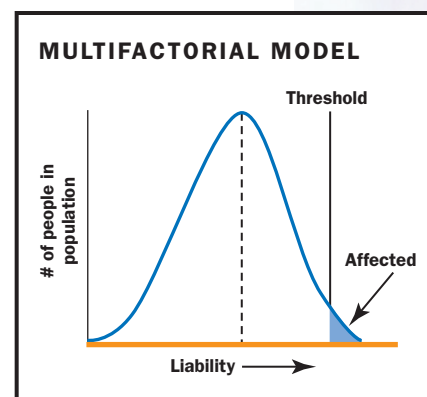


Figure 1. The threshold model for multifactorial traits. The vertical axis represents the number of people with a given level of overall liability. On the horizontal axis, individuals below the threshold do not have the disease, while those above it do.

Thawiphop Phimthep waits for his surgery to fix his cleft lip and palate in Si Saket Hospital in Sisaket Province, Thailand. Thawiphop's surgery will be performed by a group of plastic surgeons from Bangkok.



brain and spinal cord. Failure of the neural tube to close, which normally occurs during in the fourth week of gestation, results in an NTD, usually spina bifida or anencephaly. Spina bifida (“open spine”) is a defect of the spine. The most common form of spina bifida causes some degree of leg paralysis, impaired bladder and bowel control, and sometimes mental retardation. Anencephaly is a rapidly fatal condition in which a baby is born with a severely underdeveloped brain and skull.

While most NTDs are inherited as multifactorial disorders, a few result from single-gene disorders, chromosomal abnormalities, or teratogens. NTDs currently have an incidence of 1 per 2,000 births. This rate has fallen dramatically over the past thirty years, due to the remarkable effects of NTD-prevention efforts. Maternal deficiency in folate (a B vitamin) greatly increases the risk of NTDs, but taking multivitamins containing folic acid before conception and early in pregnancy is highly effective in preventing these disorders. High doses of folic acid are needed to help protect the fetuses of women with pregnancies previously affected by NTDs, and for those who need to take certain medications that interfere with folate **metabolism**.

metabolism chemical reactions within a cell

Chromosome Disorders

Humans normally have twenty-two pairs of **autosomes** and two sex chromosomes, XX or XY, making forty-six chromosomes in total. Chromosomal abnormalities occur in about 0.5 percent of all live births and are usually due to an abnormal number of chromosomes. These are nearly always an addition or deletion of a single autosome or sex chromosome in a pair. One extra copy of a chromosome is called a trisomy, while one missing copy of a chromosome is called a monosomy. Sometimes only a segment of a chromosome is duplicated or lost.

Chromosomal disorders are diagnosed by **karyotype** analysis and can be done on adults by testing blood, skin, or other tissue. Karyotypes can also be performed on a fetus through specialized testing such as amniocentesis. Prenatal maternal blood tests are routinely used to screen for some trisomies, though accurate diagnosis requires fetal karyotyping.

Chromosomal abnormalities can occur in offspring of mothers of all ages, but the frequencies of these disorders increase with maternal age, rising exponentially after the maternal age of thirty-five. Advanced paternal age has far less impact. Chromosomal abnormalities can result in either physical or functional birth defects. The severity of these birth defects is highly variable and depends upon the exact chromosome problem.

Chromosomal defects include such problems as Down syndrome, Klinefelter's syndrome, and Turner's syndrome. The majority of Down syndrome cases are due to an extra chromosome 21. Trisomy 21 usually occurs as an isolated event within a family. It results in characteristic facial features, lax muscle tone, cardiac and intestinal anomalies, and mild or moderate mental retardation. Additional medical complications may also include recurrent ear and respiratory tract infections, vision problems, hearing difficulties, and short stature.

Klinefelter's syndrome is a sex chromosome abnormality that occurs in 1 of 600 males, with a karyotype of 47. Individuals with Klinefelter's syndrome possess an extra X chromosome: XXY. Clinical characteristics are variable and include some learning and developmental disabilities, **hypogonadism**, small testes, and **gynecomastia** occurring in puberty. The condition can be managed by administering testosterone supplements beginning in adolescence. As with some other sex chromosome abnormalities, adults with Klinefelter's syndrome are usually infertile.

Turner's syndrome is another sex chromosome disorder, with a karyotype of 45. In this condition, one X chromosome is missing. Turner's syndrome occurs in one out of 4,000 live births. Most females with Turner's syndrome are short and have webbing of the neck, a broad chest, and a lack of ovarian development, with a consequent lack of pubertal development and infertility. Female hormone therapy is often used to induce breast development and menstruation. The majority of conceptions resulting in a fetus with this condition end in a miscarriage, as Turner's syndrome is highly lethal in early fetal development.

Teratogen Exposure

A teratogen is any agent that can cause birth defects if a fetus is exposed to it. Teratogens are usually drugs or infectious agents such as bacteria or

autosomes chromosomes that are not sex determining (not X or Y)

karyotype the set of chromosomes in a cell, or a standard picture of the chromosomes

The average IQ of an individual with Down syndrome is 55. Most tests cite an IQ of 100 as "average."

hypogonadism underdeveloped testes or ovaries

gynecomastia excessive breast development in males

Clubbed feet (*talipes equinovarus*) result from abnormal development of the muscles, tendons and bones of the feet in utero. Heredity is thought to play a role when such deformities appear.



viruses, and can affect a fetus from as early as the first few weeks after conception through the second trimester. For this reason, ideally, women should avoid all medications during pregnancy. However, in some circumstances there are medical risks and benefits that must be weighed, particularly if a medication is important to the health of the prospective mother.

To properly assess the potential danger of a teratogen, information is required about its effect on embryonic development, its ease of passage across the placenta, and the dosage and timing of fetal exposure to the teratogen. There are, however, limitations on the ability of doctors to predict the risk of birth defects arising from fetal exposure to any particular drug because of the lack of information on the effects of multiple medication use and possible drug interactions, the inability to control for other exposures women may have during pregnancy, and the unique genetic susceptibilities of each person. In addition, there are limited clinical studies that address this problem.

Risks for birth defects or adverse pregnancy outcomes associated with any type of exposure are in addition to the 3 percent background risk for birth defects in all pregnancies. There is no evidence linking paternal exposures to teratogenicity for the developing fetus, though exposure to some agents can reduce male fertility.

One well-documented teratogen is the drug thalidomide, which was taken by tens of thousands of women in the 1950s and early 1960s to treat nausea during pregnancy, before its potent **teratogenic** effects were recognized. Even a single dose caused severe birth defects, including amelia (absence of limbs), phocomelia (short limbs), incomplete or absent bone growth, ear and eye abnormalities, congenital heart defects, and others.

Another potent teratogen is a substance called isotretinoin, marketed under the brand name of Accutane, used to treat severe acne. Birth defects following prenatal exposure include serious central nervous system defects such as hydrocephalus, **microcephaly**, and mental retardation, as well as

teratogenic causing birth defects

microcephaly reduced head size

cleft lip and palate and anomalies of cardiovascular, limb, eye, and other systems. For women who have taken this substance, it is recommended to delay pregnancy at least one month after they have stopped using it.

Like medications, “recreational” drugs such as alcohol and cocaine can act as teratogenic agents. Fetal alcohol syndrome (FAS) and fetal alcohol effects (FAE) are the most common, completely preventable, and potentially devastating disorders caused by alcohol use during pregnancy. FAS is one of the leading causes of mental disabilities in children. Fetal exposure to alcohol creates additional problems in children born with FAS/FAE, such as characteristic facial features, growth retardation, central nervous system difficulties, learning disabilities, and behavioral problems. No amount of alcohol is thought to be safe during pregnancy; however, some of its effects may be prevented by stopping the exposure during or shortly after the first trimester.

Cocaine use during pregnancy is known to increase the risk of miscarriages and premature labor and delivery. Disturbances in the behavior of exposed newborns have been reported, such as irritability, irregular sleeping patterns, muscular rigidity, and poor feeding. Some birth defects associated with the use of this drug include urinary and genital malformations, as well as defects of the limbs, intestines, and the skull.

Conditions arising from infectious teratogenic agents include toxoplasmosis, syphilis, and rubella. In each of these cases, the mother is exposed to the infectious agent, then transmits it to the fetus. In toxoplasmosis, the parasite *Toxoplasma gondii* can be transmitted from cats to humans through contact with cat feces (cleaning litter box or gardening), or through consumption of undercooked meats, poorly washed fruits and vegetables, goat’s milk, or raw eggs. Mother-to-fetus transmission is more likely if maternal infection occurs in the last few weeks before delivery, but early fetal exposure is generally associated with greater severity of defects in the child. Overall, 20 to 30 percent of untreated, infected newborns have birth defects, including seizures, microcephaly, and other severe effects on the nervous system. Treatment of the mother with antibiotics during pregnancy is safe for the fetus, and significantly reduces the likelihood of fetal infection.

Syphilis is an infection caused by the spirochete *Treponema pallidum*. This bacterium crosses the placenta and may result in fetal infection. If untreated, the pregnancy may end in miscarriage, stillbirth, or neonatal death. Signs of congenital infection include jaundice, joint swelling, rash, anemia, and characteristic defects of bone and teeth. Maternal treatment of this condition may help prevent the transmission to the fetus and its ill effects.

Rubella is the scientific name for the disease commonly known as German measles. **Congenital** rubella syndrome (CRS) results from the exposure of an unprotected pregnant woman to the rubella virus, and can lead to major birth defects, including serious malformations of the heart, blindness, deafness, and mental retardation. CRS has been virtually eradicated in the United States because of the near-universal vaccination against rubella, now part of the standard childhood immunizations program. Unfortunately, this vastly improved situation is not as common in much of the rest of the world.

congenital from birth



enzyme a protein that controls a reaction in a cell

serum fluid portion of the blood (plural sera)

agglutinate clump together

glycoprotein protein to which sugars are attached

Maternal Conditions

Birth defects can also result from physical conditions affecting the health of the mother. One common maternal condition associated with birth defects is diabetes mellitus, a multifactorial disorder. Mothers with diabetes have a two- to three-fold times greater risk of having a child with birth defects than the general population if their condition is not well controlled. However, good glucose control has been shown to correlate with a decreased risk of congenital malformations. Characteristic diabetic malformations include cardiovascular, craniofacial, genitourinary, gastrointestinal, and neurological abnormalities. The risk that the child born of a mother with diabetes mellitus will also develop diabetes as an adult is 1 to 3 percent.

Another maternal condition giving rise to birth defects is maternal phenylketonuria (PKU). This is an autosomal recessive disorder in which an **enzyme** called phenylalanine hydroxylase is defective. This enzyme normally converts a substance in the blood called phenylalanine to another substance called tyrosine. As a result, phenylalanine levels are high, resulting in mental retardation, microencephaly, growth retardation, cardiac problems, seizures, vomiting, and hyperactivity. Other traits associated with PKU are fair hair and skin and blue eyes. PKU can be effectively managed through changes in the diet, and women who have appropriately managed their diet can have pregnancies with healthy offspring. SEE ALSO CHROMOSOMAL ABERRATIONS; CLINICAL GENETICIST; COMPLEX TRAITS; DIABETES; DOWN SYNDROME; FRAGILE X SYNDROME; GENETIC COUNSELING; GROWTH DISORDERS; SEVERE COMBINED IMMUNE DEFICIENCY; TRIPLET REPEAT DISEASE.

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Internet Resource

National Organization of Rare Disorders. <<http://www.rarediseases.org>>.

Blood Type

Blood has two main components: **serum** and cells. In 1900 Karl Landsteiner, a physician at the University of Vienna, Austria, noted that the sera of some individuals caused the red cells of others to **agglutinate**. This observation led to the discovery of the ABO blood group system, for which Landsteiner received the Nobel Prize. Based on the reactions between the red blood cells and the sera, he was able to divide individuals into three groups: A, B, and O. Two years later, two of his students discovered the fourth and rarest type, namely AB.

Antigens and Antibodies

To understand blood typing, it is necessary to define antigen and antibody. An antigen is a substance, usually a protein or a **glycoprotein**, which, when injected into a human (or other organism) that does not have the antigen,

RELATIONSHIPS BETWEEN BLOOD TYPES AND ANTIBODIES

Blood Type	Antigens on Red Blood Cell	Can Donate Blood To	Antibodies in Serum	Can Receive Blood From
A	A	A, AB	Anti-B	A, O
B	B	B, AB	Anti-A	B, O
AB	A and B	AB	None	AB, O
O	None	A, B, AB, O	Anti-A and anti-B	O

will cause an **antibody** to be produced. Antibodies are a specific type of immune-system proteins known as immunoglobulins, whose role is to fight infections by binding themselves to antigens. In the case of the ABO blood groups, the antigens are present on the surface of the red blood cell, while the antibodies are in the serum. These antibodies are unique to the ABO system and are termed “naturally occurring antibodies.” The table shows the relationships between blood types and antibodies.

This aspect of the ABO blood group system is very important in transfusion. Blood group O individuals are said to be universal donors, because their blood can be used for transfusion in individuals who have any one of the four blood types. On the other hand, individuals with blood type A can only donate to either type A or type AB, and individuals with blood type B can only donate to B or AB types. AB individuals can only donate to type AB. However, before any transfusions, donor blood is mixed with serum from the recipient (a process called cross matching) to ensure that no agglutination will occur after transfusion.

Multiple Alleles

The genetic basis of the ABO blood group system is an example of multiple **alleles**. There are three alleles, A, B, and O, at the ABO **locus** on chromosome 9. The expression of the O allele is recessive to that of A and B, which are said to be co-dominant. Thus, the genotypes AO and AA express blood type A, BO and BB express blood type B, AB expresses blood type AB, and OO expresses blood type O. In the past, ABO blood group typing was used extensively both in forensic cases as well as for paternity testing. More recently, DNA testing, which is much more informative, has superseded these tests.

The ABO blood group substances are glycoproteins, the basic molecule of which is known as the H substance. This H substance is present in unmodified form in individuals with blood type O. Adding extra sugar molecules to the H substance produces the A and B substances. The frequency of the ABO blood types varies widely across the globe. For example, blood group B has a frequency of 25 percent in Asians, 17 percent in Africans, but only 8 percent in Caucasians. The frequency of blood group O in Europe increases as one travels from southern to northern countries.

Alleles at a locus independent of the ABO blood group locus, known as the secretor locus, determine an individual's ability to secrete the ABO blood group substances in saliva and other body fluids. There are two genes, *Se* and *se*, where *Se* is **dominant** to *se*. In other words, an individual with at

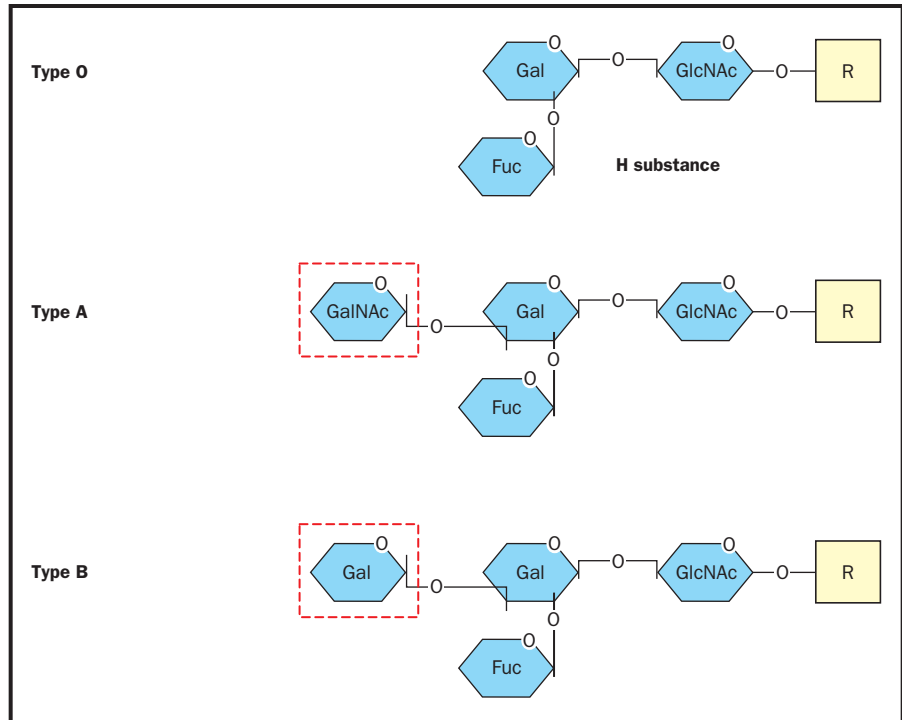
antibody immune-system protein that binds to foreign molecules

alleles particular forms of genes

locus site on a chromosome (plural, loci)

dominant controlling the phenotype when one allele is present

Molecular structure of blood type. A, B, and O antigens differ in the presence and type of the terminal sugar on a common glycoprotein base. The genes for A, B, and O blood type code for enzymes that add these sugars. Adapted from <<http://www.indstate.edu/thcme/mwking/abo-bloodgroups.gif>>.



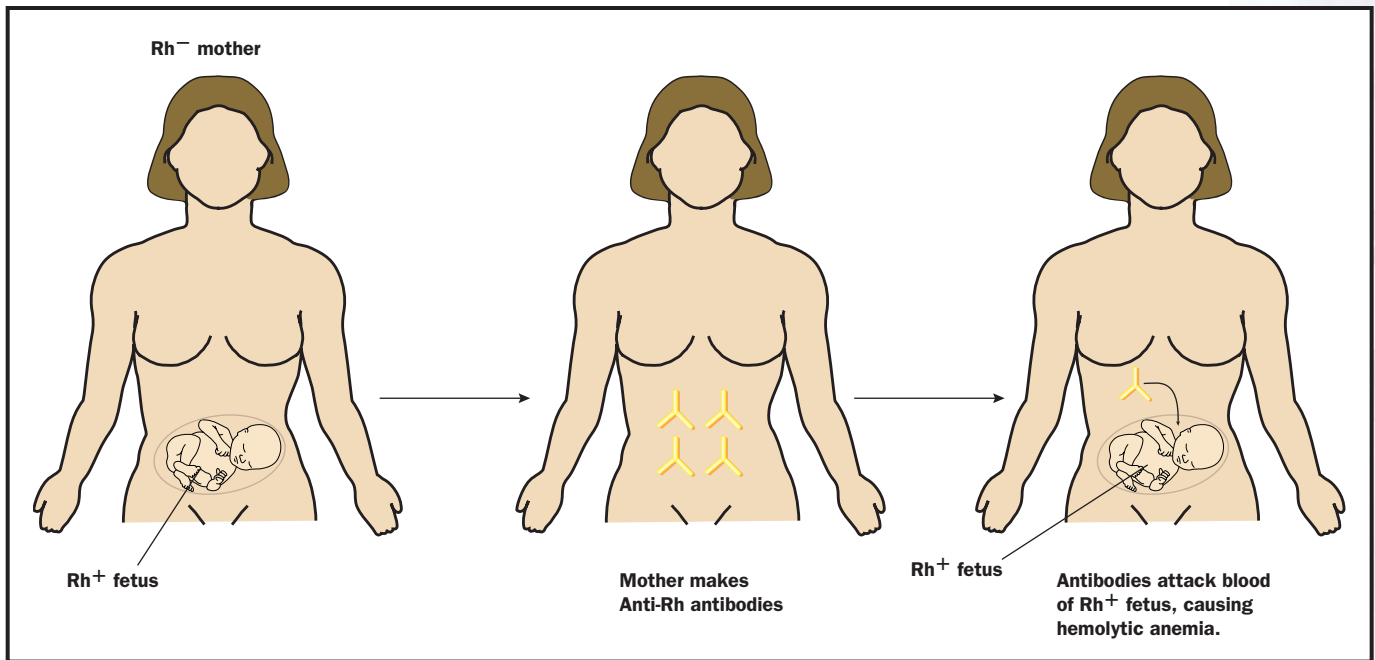
least one *Se* gene is a secretor. Approximately 77 percent of Europeans are secretors. This frequency is rarely less than 50 percent and sometimes as high as 100 percent in other populations.

An interesting aspect of the ABO blood groups is their association with disease. Among individuals with stomach and peptic ulcers, there is an excess of type O individuals, whereas among those with cancer of the stomach, there is an excess of type A individuals. Not all type O individuals have an increased risk for peptic or stomach ulcers, however. If type O individuals are secretors, they are protected against ulceration, whereas non-secretors have a two-fold increased risk. Thus the presence of ABO blood group substances act as a protective agent against the development of stomach and peptic ulcers.

The Rh System

The second most important blood group in humans is the Rhesus (Rh) system. Landsteiner and Wiener discovered the Rh blood group in 1940. They found that when they injected rabbits with Rhesus monkey blood; the rabbits produced antibodies against the Rhesus red cells. These antibodies reacted with red blood cells taken from 85 percent of Caucasians in New York City, who were thus said to be Rh positive, while the remaining 15 percent were Rh negative.

One year earlier (1939), Levine and Stetson published a paper describing the mother of a stillborn infant who had a severe reaction when transfused with her husband's blood. They tested the woman's serum and found that it reacted with 77 percent of blood donors. They postulated that the mother had been exposed to blood from her fetus and produced an antibody that reacted with it. The same antigen was present in the baby's father, explaining the woman's reaction to his blood. Their con-



clusion was correct, and later they realized that they had discovered the same antigen (Rh) that was discovered in the following year. The antibody found in the mother of the stillborn child was shown to be identical to the anti-Rh antibody produced in the rabbit by Landsteiner and Wiener.

The Rh blood group system is the major cause of **hemolytic anemia** in the newborn. A fetus who is Rh⁺ and whose mother is Rh⁻ is at high risk for this disorder, because the mother will produce antibodies against the fetal antigen. The first such fetus is usually not at risk since the fetal cells do not enter the mother's circulation until the time of birth. Only at this time does the mother produce anti-Rh⁺ antibodies. This complicates future pregnancies, because her antibodies will enter the fetal circulation system and react with fetal blood, causing **hemolysis**.

A treatment for Rh⁻ women at risk to have an Rh⁺ fetus is now widely used. Anti-Rh⁺ antibody is injected into the mother soon after her first delivery. This antibody coats the fetal Rh⁺ cells in the mother's circulation, which prevents them from causing antibody production in the mother and, therefore, her next child will not be at risk for hemolytic anemia.

The precise genetics of the complex Rh system has been in dispute since the early discoveries. The Rh blood group system is, in fact, much more complex than simply Rh⁺ and Rh⁻. There are two genes, one of which has four possible alleles, giving six antigens of which five are commonly tested. The first is D, which is the dominant gene that determines whether one is Rh⁺ or Rh⁻. Individuals with genotypes DD and Dd are Rh⁺ and those who are dd are Rh⁻. The DD and Dd genotypes cannot be distinguished from one another, since there is no "anti-d" antibody. The remaining four antigens are C, c, E, and e. The Rh locus is on the short arm of chromosome 1 and consists of two tandem genes. The first, RHCE, codes for non-

Incompatibility in Rh type can cause hemolytic anemia in a second child.

hemolytic anemia
blood disorder characterized by destruction of red blood cells

hemolysis breakdown of the blood cells

polypeptide chain of amino acids

RhD proteins while the second codes for the RhD protein. The Rh **polypeptide** has been sequenced. It contains 417 amino acids. Thus the molecular genetics conferring different antigenic Rh types is now clear. SEE ALSO GENOTYPE AND PHENOTYPE; IMMUNE SYSTEM GENETICS; INHERITANCE PATTERNS; MULTIPLE ALLELES.

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Blotting

Blotting is a common laboratory procedure in which biological molecules in a gel matrix are transferred onto nitrocellulose paper for further scientific analysis. The biological molecules transferred in this process are DNA fragments, RNA fragments, or proteins. Because the isolation and characterization of these types of materials is at the center of much molecular biology research, blotting is one of the most useful techniques in the molecular biology laboratory.

The blotting procedure is named differently depending on the type of the molecules being transferred. When the molecules to be transferred are DNA fragments, the procedure is called a **Southern blot**, named for the man who first developed it, Edward Southern, a molecular biologist at Oxford University. The **Northern blotting** procedure, which transfers RNA molecules, was developed shortly thereafter and, since it was patterned after Southern blotting, its name was a humorous play on words inspired by the name of the first procedure. Western blotting got its name in a similar fashion. All three blotting methods are relatively easy to carry out, can be conducted in a short period of time, and provide answers to many questions that are commonly raised in the field of molecular biology.

The Procedure

All blotting procedures begin with a standard process called **gel electrophoresis**. During this step, DNA, RNA, or proteins are loaded on to an agarose or acrylamide gel (that functions like a molecular sieve) and are then run through an electric field. Two types of gels are commonly used: agarose gels and acrylamide gels. Agarose gels are based on a meshwork of agar filaments and are most often used to analyze DNA and RNA. Acrylamide gels are based on a meshwork formed from the chemical acrylamide and used most often to analyze proteins. Gels are loaded with a mixture of many differently sized molecules. When pulled through the

Southern blot a technique for separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe

Northern blotting separating RNA molecules by electrophoresis and then identifying a target fragment with a DNA probe

gel electrophoresis technique for separation of molecules based on size and charge

gel by the electric current, they will separate into separate pools on the basis of their size; smaller molecules migrate farther through the gel than larger molecules. These separate pools of molecules will appear as bands on the gel if they are stained with an appropriate dye. After the molecules have been fractionated on the gel, they are ready for transfer to the nitrocellulose paper.

Transfer is initiated when the gel is retrieved from the electrophoresis apparatus and the nitrocellulose paper is carefully laid on top of the gel. The objective now is to transfer the bands of molecules found in the gel over to the nitrocellulose paper. Here they become immobilized, and will reflect the pattern seen on the gel. The paper now serves as a type of permanent record of the gel's banding pattern that can be used for further analysis.

There are two basic ways the actual transfer, or blotting, is carried out. One method takes a “sandwich” of gel and nitrocellulose paper and places it in a special apparatus that sets up an electric field running perpendicular to the band as preserved in the gel. This pulls the bands of molecules out of the gel, and they are immediately absorbed onto the nitrocellulose paper. This method is most commonly employed in Western (protein) blots.

The other method, commonly employed with Southern and Northern blots, lays the gel on top of a platform that in turn is placed in a tray containing a buffer solution. Underneath the gel is a strip of blotting paper that is folded down on each side of the platform, so that it dips into the buffer to serve as a wick. On top of the gel are placed, first, the strip of nitrocellulose paper, then several pieces of blotting paper, and finally a small stack of paper towels. A weight is then placed on top of the paper towels. The buffer flows up the blotting paper “wick” by capillary action, then through the gel, through the nitrocellulose paper, and ultimately into the paper towels. The DNA or RNA in the gel moves with the buffer but sticks to the nitrocellulose paper on contact. The paper towels soak up the transfer buffer, but only after it has passed through the gel and deposited the DNA or RNA on the nitrocellulose paper.

After transfer has been completed, the nitrocellulose paper can be examined by using probes. Short fragments of DNA that have a **nucleotide** sequence complementary to the molecule being analyzed are normally used as probes in Southern and Northern blots. **Antibodies** that react with the protein being analyzed are used as probes in a Western blot. In either case, the probe is “labeled,” usually by making it radioactive, so that it is easy to identify. In all blotting experiments, the nitrocellulose paper is placed in a chamber full of buffer and mixed with the probe, which then binds to the molecule that is being studied. This is called the **hybridization** step. Detection of the probe indirectly detects the molecules being studied.

Illustrative Examples

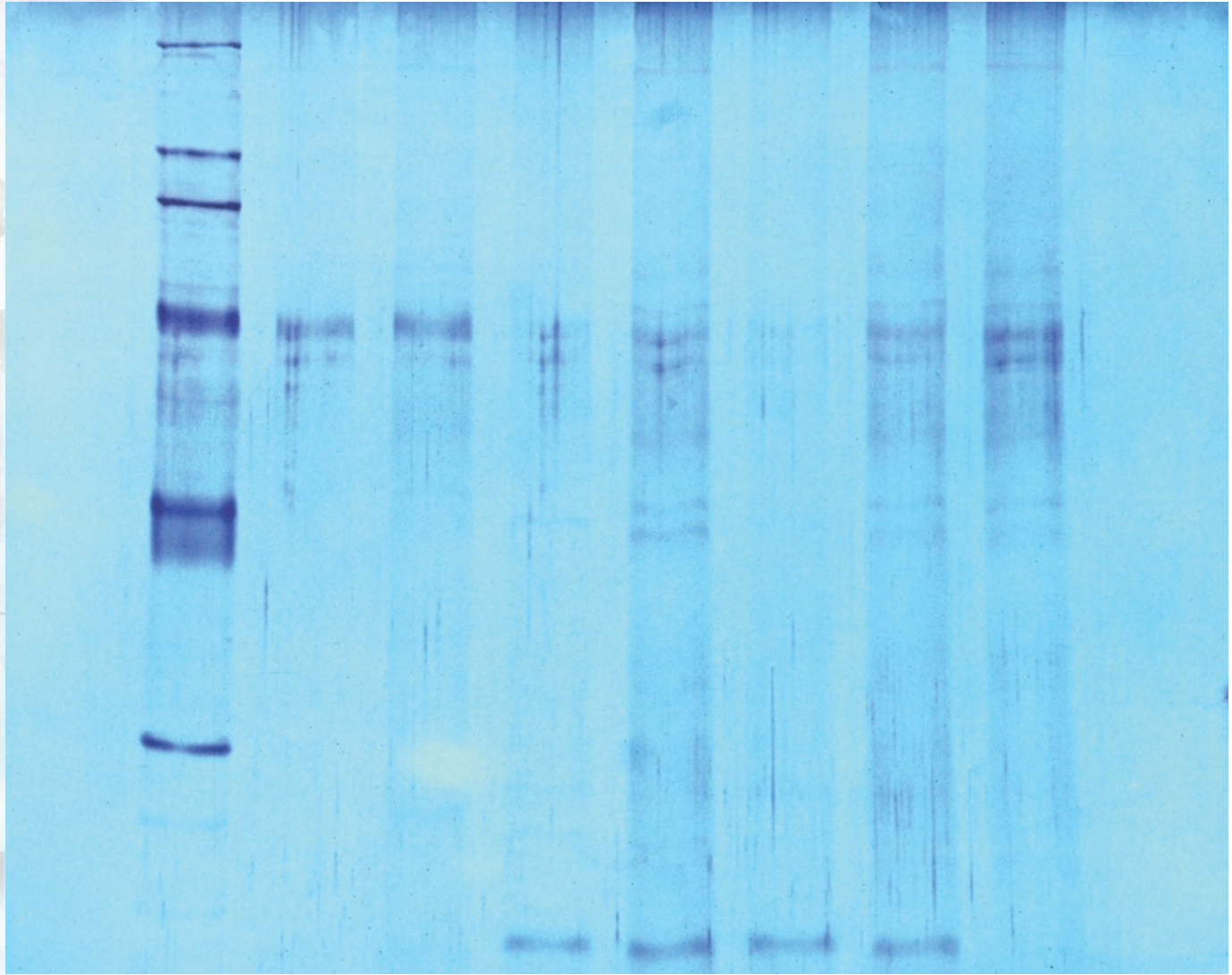
Blotting is perhaps best understood with illustrative examples. Suppose a student was studying a newly identified gene, X, from cows. The student then asks three basic questions as part of a research project: (1) Do pigs also have gene X on their chromosomes? (2) Do cows express gene X in their brain tissue? (3) Is the protein product of gene X found in the cow's blood plasma? Blotting experiments can answer all three of these questions.

nucleotide the building block of RNA or DNA

antibodies immune-system proteins that bind to foreign molecules

hybridization (molecular) base-pairing among DNAs or RNAs of different origins





Southern blots, such as this one, are useful in forensic examinations of DNA evidence.

restriction enzyme an enzyme that cuts DNA at a particular sequence

probe molecule used to locate another molecule

complementary matching opposite, like hand and glove

A Southern (DNA) blot will answer the first question. The student obtains DNA from a pig, uses a **restriction enzyme** to cut the DNA into a large pool of fragments of different sizes, and then fractionates the DNA fragments using gel electrophoresis. The contents of the gel are then chemically treated so that the double-stranded DNA molecule “unzips” and exists in a single-stranded form, which is then blotted onto nitrocellulose paper. At this point, the student can take gene X (or a portion of the gene) from the cow, label it, make it single-stranded, and use it as a **probe** to analyze the pig’s DNA. The labeled probe is then added to the nitrocellulose blotted with the pig DNA. If the pig’s DNA also contains gene X, there should be a fragment on the nitrocellulose with a nucleotide sequence sufficiently **complementary** to the probe such that the probe will bind. In other words, the labeled probe will bind to any fragment from the blotted pig DNA that contains gene X, allowing the student to detect the presence of gene X in pigs.

To answer the second question, a Northern (RNA) blot would be used. The procedure is essentially the same as with the Southern blot, except that

the student would isolate RNA from the cow's brain tissue and run it out on the gel. The same DNA probe described above would then be used to detect whether the RNA that represents gene X expression is present in the brain.

To answer the third question, the student would use a Western (protein) blot. This requires the use of an antibody that specifically reacts with the protein coded for by gene X. The student first obtains plasma from the cow and uses standard biochemical techniques to isolate the proteins for analysis. These proteins can then be run out on a gel and transferred to nitrocellulose. The proteins can then be probed with the labeled antibody. If the product of gene X is in the plasma, it will bind with the labeled antibody and can thus be detected. **SEE ALSO** GEL ELECTROPHORESIS; *IN SITU* HYBRIDIZATION; RESTRICTION ENZYMES; SEQUENCING DNA.

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Breast Cancer

Breast cancer remains the most common cause of cancer among women in the United States, and it results in more deaths from cancer among women than any other type of cancer, except lung cancer. Over 40,000 women die from breast cancer in the United States each year. A long history of research, now coupled with the new information emerging from the field of molecular genetics, is beginning to explain the basic steps leading to breast cancer, and it will enable the development of novel treatment and prevention strategies.

Almost all breast cancers begin in the glandular structures in the breast that, during lactation, produce milk. These mammary glands are under the control of reproductive hormones that stimulate the monthly cycle of gland expansion and shrinkage, which is a feature of the regular menstrual cycle. Many of the factors associated with the development of breast cancer appear to have their effect through interaction with the hormonal stimulation of these glands.

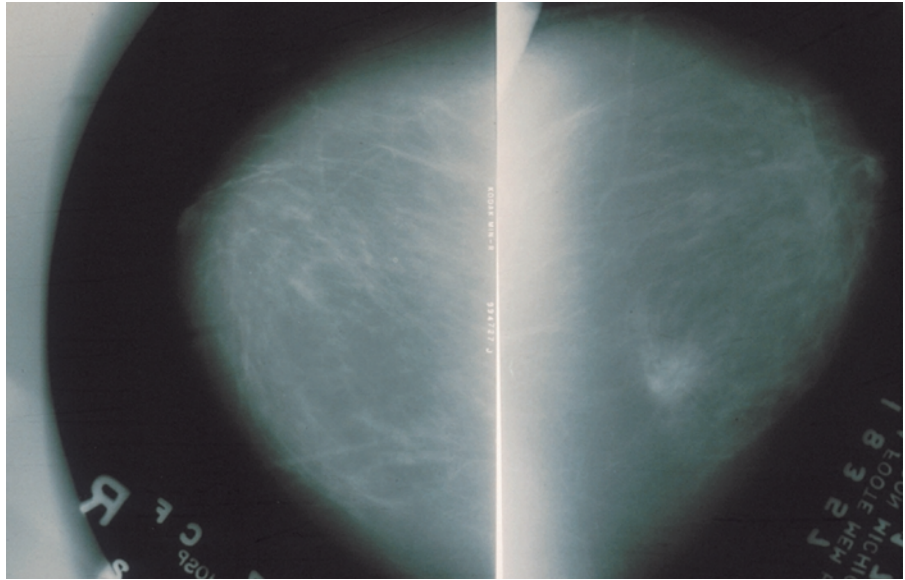
The risk of developing breast cancer increases throughout a woman's lifetime, and the disease is relatively rare in very young women. The overall association of breast cancer incidence with increasing age may be explained by a model of breast cancer in which a progressive and cumulative series of genetic changes within the cells of the glands is necessary for the initiation of cancer. The longer a woman lives, the more opportunities there are for these genetic changes to accumulate and reach a stage where cells can become cancerous.

One of the most consistent epidemiological observations is the association of reproductive events with risk of breast cancer. Women who have

MALE BREAST CANCER

According to the National Cancer Institute, male breast cancer is most common among males between 60 and 70 years of age. Two of the major risk factors for men include: exposure to radiation, and having a family history of breast cancer (especially the *BRCA2* gene). The survival rate for men with breast cancer almost equals that for women.

This composite of two mammograms compares healthy breast tissue (left) to partly cancerous breast tissue (right). The malignant lump is visible as a lighter patch in the mammogram on the right.



one or more full-term pregnancies have a *lower* risk for breast cancer, especially if they are pregnant before age twenty. Pregnancy at an early age may help to stabilize the mammary glands and make them less vulnerable to genetic changes later in life. The risk for breast cancer is also significantly decreased among women undergoing surgical removal of the ovaries, particularly if the surgery is performed before age thirty-five. This surgery removes the major source of reproductive hormones and therefore results in less stimulation of the glands in the breast.

Conversely, the greater number of years a woman has regular menstrual cycles, the higher the risk of breast cancer. There is also a modest increase in risk associated with postmenopausal estrogen replacement therapy (especially when used more than 15 years), and with exposure to the synthetic estrogen diethylstilbestrol during pregnancy. Studies have found a significant correlation between breast cancer and levels of hormones—estradiol, estrone, estrone sulfate, prolactin, and dehydroepiandrosterone sulfate. A drug used to treat breast cancer, tamoxifen, blocks **estrogen** receptors.

Taken together, a significant body of research shows that reproductive hormones—produced internally and taken as medicines—are major determinants of breast cancer risk. Other factors—including genetic predisposition, environmental exposure, and lifestyle choices—may increase cancer risk via hormone regulation.

There are striking racial and ethnic differences in breast cancer incidence and resulting deaths. Overall, rates are highest for Caucasian women and lowest for Native American and Korean women. The general international pattern of breast cancer incidence reveals higher rates for Western, industrialized nations, and lower rates for less industrialized and Asian countries. Even within the United States, there is significant geographic diversity in breast cancer rates, with mortality rates highest in the Northeast and lowest in the South. Much of this variation is thought to be due to regional differences in reproductive events, such as the age when women start having children and their use of hormone medications.

estrogen female hormone

There is also considerable evidence from international comparisons, migration studies, and time trends to support an important role for dietary fat in the causation of breast cancer. However when the diets of specific population groups are followed over time, no definite causal link can be demonstrated. The data on fiber and vitamins and minerals is also contradictory. Dietary studies also show a fairly consistent but weak increase in breast cancer risk with moderate to heavy alcohol consumption. Alcohol may act by stimulating the production of more internal hormones. Among postmenopausal women, body weight has also been positively correlated with both breast cancer incidence and mortality. Although exposure to large amounts of radiation is associated with an increased risk for breast cancer, there does not appear to be any risk associated with routine diagnostic imaging, such as chest X rays and mammograms.

Finally, there is limited data to support a protective role for physical activity, both during leisure time and at work, in terms of breast cancer risk. The effect is most pronounced among premenopausal and younger postmenopausal women. The known association of vigorous physical activity with decreased circulating levels of ovarian hormones may explain this finding, which could have significant public health implications.

Women undergoing breast **biopsies** whose tissue shows no evidence of cancer, but whose cells have **atypical** features or faster-than-normal rates of growth have an increased risk of breast cancer, with risks up to eightfold higher in some cases. It is thought that these atypical cells may be a precursor to the development of breast cancer, or they may act as markers for genetic instability within the glandular cells.

Population studies have documented that a history of breast cancer in first-, second-, or third-degree relatives increases cancer risk between twofold and fourfold. Recently two genes, *BRCA1* and *BRCA2*, have, when inherited in a mutated form, been associated with a hereditary form of breast cancer. This form is characterized by early age at onset (5 to 15 years earlier than noninherited cases), cancer in both breasts, and association in the family with tumors of other organs, particularly of the ovary in women and prostate gland in men. Among the normal functions of these genes are the control of the cell cycle and the maintenance of stability of the genes. Both genes are tumor suppressor genes whose proteins help both to control the cell cycle and to repair damaged DNA. Mutations interfere with this vital function, causing damaged cells to reproduce and become cancerous.

The frequency of mutations in *BRCA1* in the general population has been estimated to be 1 in 800. Carrier rates are not distributed evenly, however, and mutations tend to concentrate in families with multiple cases of breast or ovarian cancer. Different ethnic groups have unique *BRCA1* and *BRCA2* mutations. Most notably, three specific mutations are common in Ashkenazic Jews. Additional **founder** mutations have been described in Sweden and Iceland.

Individuals who have inherited a mutated *BRCA1-2* gene face an estimated 36 percent to 85 percent lifetime risk for breast cancer and an estimated 16 percent to 60 percent lifetime risk for ovarian cancer. Among female *BRCA1* carriers who have already developed a primary breast cancer, estimates for a second breast cancer in the opposite breast are as high as 64 percent by age seventy. Men who test positive for a mutation in the *BRCA2* gene also have a higher lifetime risk for breast cancer.

biopsies removal of tissue samples for diagnosis

atypical irregular

founder population

The identification and location of these breast cancer genes will now permit further investigation of the precise role they play in cancer progression and, specifically, how they interact with reproductive hormones. SEE ALSO CANCER; CELL CYCLE; COLON CANCER; ONCOGENES; TUMOR SUPPRESSOR GENES.

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Caenorhabditis elegans See *Roundworm: Caenorhabditis elegans*

Cancer

Cancer is a number of related diseases that are characterized by the uncontrolled proliferation and disorganized growth of cells. Tumor cells invade and destroy normal tissues and may spread throughout the body via the circulatory systems.

A Genetic Disease

Cancer is the result of changes in the genetic material of a cell that cause the cell to gradually lose the ability to grow in a regulated fashion. These changes can be brought about by contact with harmful environmental agents or by inheritance of genes leading to a **genetic predisposition**.

Cancer risk increases with age, as the probability of accumulating **mutations** in the DNA increases with time. Environmental factors include lifestyle (e.g., smoking), diet (e.g., saturated fats from red meat), and exposure to certain chemicals (e.g., asbestos, benzopyrenes), ionizing radiation (e.g., X-rays, radon gas), ultraviolet radiation (e.g., sun, tanning beds), and certain viruses (e.g., human papillomavirus, Epstein-Barr virus). Heredity also plays a role in **oncogenesis**, as mutations in certain genes increase the probability of developing certain types of cancer. For instance, women who inherit a mutated copy of the *BRCA1* or *BRCA2* gene have a greatly increased probability of developing breast cancer at a young age.

Classification of Cancer Types

The term "cancer" is general, in that it represents a large group of related diseases that arise from **neoplasms**. A neoplasm is classified by the type of tissue in which it arises and the stage to which it has progressed. Neoplasms are also called tumors. Not all tumors are cancerous. A tumor that grows in one place and does not invade surrounding tissue is called benign. In contrast, invasive tumors are called malignant. These are cancerous.

genetic predisposition increased risk of developing diseases

mutations changes in DNA sequences

oncogenesis the formation of cancerous tumors

neoplasms new growths